AUDIT REPORT

genai-recommender: demo video audit report

A slide-by-slide record of every number, assumption and calculation in the demo video, with its source.

Item	Value
Repository	https://github.com/kavya2693/genai-recommender
Numbers taken from commit	`37d49ecc8dcf2a405ea628ac0a26027e2d2aaa41` (outputs last changed in `37d49ec (2026-10-14 12:39:30 +0400)`)
Video	`docs/demo/demo.mp4` · 89.0 s · 1920x1080 · 30 fps · silent
Generated	2026-10-14 19:04
Automated checks	72 pass · 0 fail · 1 manual (of 73 figures)
Assumptions	14 (A-01 to A-14)

How to use this report

Every number or claim on screen has an ID (F-01, F-02, ...) and every modelling assumption has an ID (A-01, ...). Cite the ID. Each figure lists its source file, the exact column, row or key, the formula for derived numbers, and an automated check result. The change-impact matrix (section 5) shows which slides and files change if an input or assumption changes.

Types: direct = read straight from a repo output; derived = computed from direct values (formula given); illustrative = drawing or example, not data; general knowledge = context not produced by the repo.

Regenerate

```
make demo                                  # outputs/*.csv, summary.json, bert4rec_training.json,
drafts_llama3.2.md (~4 min; needs ollama pull llama3.2)
PYTHONPATH=src .venv/bin/python docs/demo/source/tools/export_outreach_plan.py
docs/demo/source/assets/outreach_plan.json --region north   # ~6 min, retrains with seed 7
python3 docs/demo/source/tools/inject_sheet.py docs/demo/source/assets/outreach_plan.json
docs/demo/source/compositions/value.html --rows 0,1,2,9,15,66
cd docs/demo/source && npx --yes hyperframes@0.8.156 check && npx --yes hyperframes@0.8.156 render -f 30 -q
looks -o renders/genai-recommender-master.mp4 && cd -
ffmpeg -y -i docs/demo/source/renders/genai-recommender-master.mp4 -c:v libx264 -preset slow -crf 25 -pix_fmt
yuv420p -movflags +faststart -an docs/demo/demo.mp4
python3 docs/demo/source/tools/build_audit_report.py docs/demo/source/audit/facts.yaml --master
docs/demo/source/renders/genai-recommender-master.mp4   # stills, checks, MD/DOCX/PDF
```

1. Figures

ID	Shown	Slides	Source	Column / row / key	How computed	Type	Check
F-01	3 chips	01	author's framing	-	-	general knowledge	manual: Context statement, not a repo output.
F-02	4 problem cards	01	README 'The problem'	README lines 13-27	-	general knowledge	✓ verified
F-03	20,000 prescribers	02	data/customers.csv	row count	-	direct	✓ verified
F-04	480 pieces of content / top-10 of 480 / all 480 items	02, 05, 09	data/items.csv	row count	-	direct	✓ verified
F-05	Email, SMS or a rep visit?	02, 03	src/genai_recommender/datagen.py:62	CHANNELS	-	direct	✓ verified
F-06	626,414 reads	02	data/interactions.csv.gz	row count	-	direct	✓ verified
F-07	over 2 years / 2 yrs	02, 05	src/genai_recommender/datagen.py:65-66; data/interactions.csv.gz ts	DAYS = 730; first and last read 2024-09-01 .. 2026-08-31	span = 729 days + 1	direct	✓ verified
F-08	299,722 logged sends	02	data/sends.csv.gz	row count	-	direct	✓ verified
F-09	252 expired claims look like current ones	02	data/snippets.csv	status == expired	count	direct	✓ verified
F-10	3 green badges	03	outputs/retrieval.csv; outputs/summary.json	arm 'approved only + brand filter': unapproved = 0, other-brand = 0; stub_grounded_share = 1.0; llm_demo grounded 3/3	-	derived	✓ verified
F-11	learns reading order	03	src/genai_recommender/bert4rec.py:105-130	_training_batch masking	-	direct	✓ verified
F-12	S19 APPROVED 47% · S22 EXPIRED 28% · S23 EXPIRED 100 mg	04	data/snippets.csv	snippet_id 19, 22, 23	-	direct	✓ verified
F-13	73%	04	outputs/retrieval.csv	arm='all statuses indexed', any unapproved claim retrieved (0.7342)	x 100, rounded	direct	✓ verified
F-14	3 step statements	04	src/genai_recommender/rag.py:36-37, :57-58, :163	-	-	direct	✓ verified
F-15	This design → 0%	04	outputs/retrieval.csv	arm='approved only + brand filter', any unapproved claim retrieved	x 100	direct	✓ verified
F-16	79 test queries / 79 claim queries	04, 08	outputs/retrieval.csv	queries	-	direct	✓ verified
F-17	RANK 0.67 ms	05	outputs/latency.csv	stage=rank, p50 ms	-	direct	✓ verified
F-18	CHANNEL 18.94 ms	05	outputs/latency.csv	stage=channel, p50 ms	-	direct	✓ verified
F-19	RETRIEVE + DRAFT 0.80 ms	05	outputs/latency.csv	stage=retrieve+draft, p50 ms	-	direct	✓ verified
F-20	95% ≤ 23.27 ms, no LLM / 23.3 ms	05, 08	outputs/latency.csv	stage=total, p95 ms	-	direct	✓ verified
F-21	626k reads · 300k sends · 489 claims	05	data/interactions.csv.gz, data/sends.csv.gz, data/snippets.csv	row counts 626,414 / 299,722 / 489	rounded to thousands	derived	✓ verified
F-22	–40 scanner bots	05	outputs/summary.json	scanners_dropped	-	direct	✓ verified
F-23	226 approved only	05	data/snippets.csv	status == approved	count	direct	✓ verified
F-24	word-based search (TF-IDF + SVD, 64-d, not shown in full)	05	src/genai_recommender/rag.py:32-44	SnippetIndex(dim=64)	-	direct	✓ verified
F-25	133k params, CPU (05) · parameters · 2 layers (06)	05, 06	outputs/summary.json	bert4rec.config.layers, dim; bert4rec.parameters = 133,345	parameters rounded to thousands	direct	✓ verified
F-26	seed 7 · 2 yrs	05	src/genai_recommender/datagen.py:455, :66	seed: int = 7; DAYS = 730	-	direct	✓ verified
F-27	top-10 of 480 · read ones removed	05	src/genai_recommender/serve.py:60-61	mask_seen, argsort[:10]	-	direct	✓ verified
F-28	5.6–6.6 s / draft	05	outputs/summary.json	llm_demo.draft_ms = [5570, 6178, 6605]	min, max / 1000	derived	✓ verified
F-29	LLM draft ≈ 300× the rest	05	outputs/summary.json; README line 149	mean draft_ms 6,118 / total p50 20.44 ms = 299	mean(draft_ms) / total p50, rounded to the nearest hundred	derived	✓ verified
F-30	llama3.2 · 3B · Ollama · CPU · stand-in for tests	05	README line 148; src/genai_recommender/rag.py StubLLM, OllamaLLM	-	-	direct	✓ verified
F-31	datagen.py, data.py, channel.py (x3),	05	src/genai_recommender/	functions and classes	-	direct	✓ verified

ID	Shown	Slides	Source	Column / row / key	How computed	Type	Check
	rag.py (x2), bert4rec.py, serve.py (x2), cli.py, baselines.py						
F-32	only past history · 1 row per channel	05	src/genai_recommender/serve.py:64-72; channel.py:46-58	-	-	direct	✓ verified
F-33	14 points, epochs 2-28	06	outputs/bert4rec_training.json history	epoch, valid_ndcg10	-	direct	✓ verified
F-34	epoch 22 · 0.303	06	outputs/bert4rec_training.json	best_epoch, best_valid_ndcg10	-	direct	✓ verified
F-35	stopped at 28	06	outputs/bert4rec_training.json	last history epoch	-	direct	✓ verified
F-36	196 s	06	outputs/summary.json	bert4rec.train_seconds	-	direct	✓ verified
F-37	4 CPU threads, no GPU	06	README line 57; cli.py --threads default min(4, cpu_count)	-	-	direct	✓ verified
F-38	133,345	06	outputs/summary.json	bert4rec.parameters	-	direct	✓ verified
F-39	576,746 training events · 19,960 prescribers	06, 08, 09	outputs/summary.json	split.train_events, split.users	-	direct	✓ verified
F-40	Prescriber 2651 · Somnavel elderly patients series, part 5 (BERT4Rec's top pick of 480, not yet read) · email 0.194 / SMS 0.114 / rep 0.416 · S21 S20 S19 · 6,605 ms	06	outputs/drafts_llama3.2.md lines 44-49	Customer 2651 section	-	direct	✓ verified
F-41	neurology · no rep coverage · rep 0.416 not eligible	06	data/customers.csv	customer_id 2651	-	direct	✓ verified
F-42	claim check: pass · all 8 numbers in S19–S21 or the title	06	outputs/drafts_llama3.2.md; data/snippets.csv; rag.py grounding_check logic	-	numbers in draft (regex \d+(\.\d+)?) all in claims + title	derived	✓ verified
F-43	3 of 3 drafts pass	06	outputs/summary.json	llm_demo.samples, grounded	-	direct	✓ verified
F-44	3,275 prescribers / 3,275 planned	07	demo assets/outreach_plan.json	prescribers	-	direct	✓ verified
F-45	395 rep visits, within budget	07	demo assets/outreach_plan.json	channels.rep = 395, rep_budget = 435	395 <= 435	direct	✓ verified
F-46	every draft passes the claim check	07	demo assets/outreach_plan.json	grounded == prescribers	3,275 / 3,275	direct	✓ verified
F-47	HCP-0001, -0002, -0003, -0010, -0016, -0067	07	demo assets/outreach_plan.json, injected into compositions/value.html	rows 0, 1, 2, 9, 15, 66	HCP id = row position + 1	direct	✓ verified
F-48	+ 3,269 more prescribers	07	F-44	3,275 - 6	-	derived	✓ verified
F-49	(supports every slide 07 row)	07	demo assets/outreach_plan.json ranking_check vs outputs/ranking.csv	HR@10, NDCG@10, coverage@10 per model	equal at 4 decimals	direct	✓ verified
F-50	+8.9%	08	outputs/channel_policy.csv	engaged per 1k touches: model 206.64 vs logged 189.77	(206.64 / 189.77 - 1) x 100 = 8.89%	derived	✓ verified
F-51	189.8 → 206.6 per 1,000	08	outputs/channel_policy.csv	engaged per 1k touches	-	direct	✓ verified
F-52	62% of a perfect plan's gain recovered	08	outputs/channel_policy.csv	policy='model, same rep budget', share of oracle lift (0.6173)	x 100	direct	✓ verified
F-53	0.0% (index everything: 73.4%)	08	outputs/retrieval.csv	any unapproved claim retrieved	x 100	direct	✓ verified
F-54	today 0% · perfect 100%	08	outputs/channel_policy.csv	share of oracle lift for logged and oracle	-	direct	✓ verified
F-55	23.3 ms · 95% of requests take this or less · request path, no LLM	08	outputs/latency.csv	stage=total, p95 ms	1 decimal	direct	✓ verified
F-56	channel: 93% of median	08	outputs/latency.csv	18.94 / 20.44	channel p50 / total p50 x 100 = 92.7%	derived	✓ verified
F-57	plans scored on the data generator's true engagement	08	src/genai_recommender/channel.py:274	docstring	-	direct	✓ verified
F-58	300 timed requests	08	outputs/latency.csv	requests	-	direct	✓ verified
F-59	one make demo run	08	README line 57; outputs/summary.json llm_demo present	-	-	direct	✓ verified
F-60	(rep budget, slide 07 'within budget')	07, 08	outputs/channel_policy.csv;	rep share of the logged policy = 0.1328 = export	-	direct	✓ verified

ID	Shown	Slides	Source	Column / row / key	How computed	Type	Check
			demo assets/outreach_plan.json	rep_share_historical			
F-61	0.316 · loses: 0.303 vs 0.316	09	outputs/ranking.csv	model='last-item Markov', NDCG@10	-	direct	✓ verified
F-62	0.303	09	outputs/ranking.csv	model='BERT4Rec', NDCG@10	-	direct	✓ verified
F-64	0.024	09	outputs/ranking.csv	model='popularity', NDCG@10	-	direct	✓ verified
F-65	Lookup 0.365 USE · BERT4Rec 0.232	09	outputs/ranking_by_segment.csv	slice='step=pathway', NDCG@10	-	direct	✓ verified
F-66	BERT4Rec 0.461 USE · Lookup 0.375	09	outputs/ranking_by_segment.csv	slice='step=series_resume', NDCG@10	-	direct	✓ verified
F-72	NDCG@10 0.331 vs 0.316 today · a ceiling, not a result	09	outputs/ranking_by_segment.csv	slices step=series, series_resume, pathway, pref (users 3,624 + 2,392 + 4,145 + 9,799 = 19,960)	sum(users x max(BERT4Rec, Markov)) / 19,960 = 0.3312; Markov alone = 0.3158 (the same weighting reproduces outputs/ranking.csv for both models)	derived	✓ verified
F-73	NEXT · NOT BUILT YET · The README's next experiment: a blend, or BERT4Rec fed the lookup's score. Not in the code yet.	09	README 'Insights and learning'; src/genai_recommender/*.py; serve.py:59	-	-	direct	✓ verified
F-74	Back to a series after other reads · Next format: article → dosing guide	09	README 'The result' slice table	-	-	direct	✓ verified
F-67	MIT	10	LICENSE line 1	-	-	direct	✓ verified
F-68	PyTorch · scikit-learn · Ollama · llama3.2 · Python	10	pyproject.toml; src/genai_recommender/rag.py	torch, scikit-learn, OllamaLLM(model='llama3.2')	-	direct	✓ verified
F-69	$ make demo	10	Makefile	demo target	-	direct	✓ verified
F-70	+ 3 llama3.2 drafts	10	Makefile demo target	--demo-n 3	-	direct	✓ verified
F-71	~4 min on CPU	10	outputs/summary.json	wall_seconds = 233.4	233.4 / 60 = 3.9 min	derived	✓ verified

2. Automated check run (2026-10-14 19:04)

ID	Expected on slide	Recomputed from repo	Result
F-01	3 chips	-	manual
F-02	true	true	✓ pass
F-03	20,000	20,000	✓ pass
F-04	480	480	✓ pass
F-05	true	true	✓ pass
F-06	626,414	626,414	✓ pass
F-07	true	true	✓ pass
F-08	299,722	299,722	✓ pass
F-09	252	252	✓ pass
F-10	true	true	✓ pass
F-11	true	true	✓ pass
F-12	true	true	✓ pass
F-13	73%	73%	✓ pass
F-14	true	true	✓ pass
F-15	0%	0%	✓ pass
F-16	79	79	✓ pass
F-17	0.67	0.67	✓ pass
F-18	18.94	18.94	✓ pass
F-19	0.80	0.80	✓ pass
F-20	23.27	23.27	✓ pass
F-21	626k 300k 489	626k 300k 489	✓ pass
F-22	40	40	✓ pass
F-23	226	226	✓ pass
F-24	true	true	✓ pass
F-25	2 64 133k	2 64 133k	✓ pass
F-26	true	true	✓ pass
F-27	true	true	✓ pass
F-28	5.6-6.6	5.6-6.6	✓ pass
F-29	300x	300x	✓ pass
F-30	true	true	✓ pass
F-31	true	true	✓ pass
F-32	true	true	✓ pass
F-33	true	true	✓ pass
F-34	22 0.303	22 0.303	✓ pass
F-35	28	28	✓ pass

ID	Expected on slide	Recomputed from repo	Result
F-36	196 s	196 s	✓ pass
F-37	true	true	✓ pass
F-38	133,345	133,345	✓ pass
F-39	576,746 19,960	576,746 19,960	✓ pass
F-40	true	true	✓ pass
F-41	true	true	✓ pass
F-42	true	true	✓ pass
F-43	3 of 3	3 of 3	✓ pass
F-44	3,275	3,275	✓ pass
F-45	true	true	✓ pass
F-46	true	true	✓ pass
F-47	true	true	✓ pass
F-48	3,269	3,269	✓ pass
F-49	true	true	✓ pass
F-50	+8.9%	+8.9%	✓ pass
F-51	189.8 → 206.6	189.8 → 206.6	✓ pass
F-52	62%	62%	✓ pass
F-53	0.0% 73.4%	0.0% 73.4%	✓ pass
F-54	true	true	✓ pass
F-55	23.3	23.3	✓ pass
F-56	93%	93%	✓ pass
F-57	true	true	✓ pass
F-58	300	300	✓ pass
F-59	true	true	✓ pass
F-60	true	true	✓ pass
F-61	0.316	0.316	✓ pass
F-62	0.303	0.303	✓ pass
F-64	0.024	0.024	✓ pass
F-65	0.365 0.232	0.365 0.232	✓ pass
F-66	0.461 0.375	0.461 0.375	✓ pass
F-72	0.331 0.316	0.331 0.316	✓ pass
F-73	true	true	✓ pass
F-74	true	true	✓ pass
F-67	true	true	✓ pass
F-68	true	true	✓ pass
F-69	true	true	✓ pass
F-70	true	true	✓ pass
F-71	~4 min	~4 min	✓ pass

3. Assumptions register

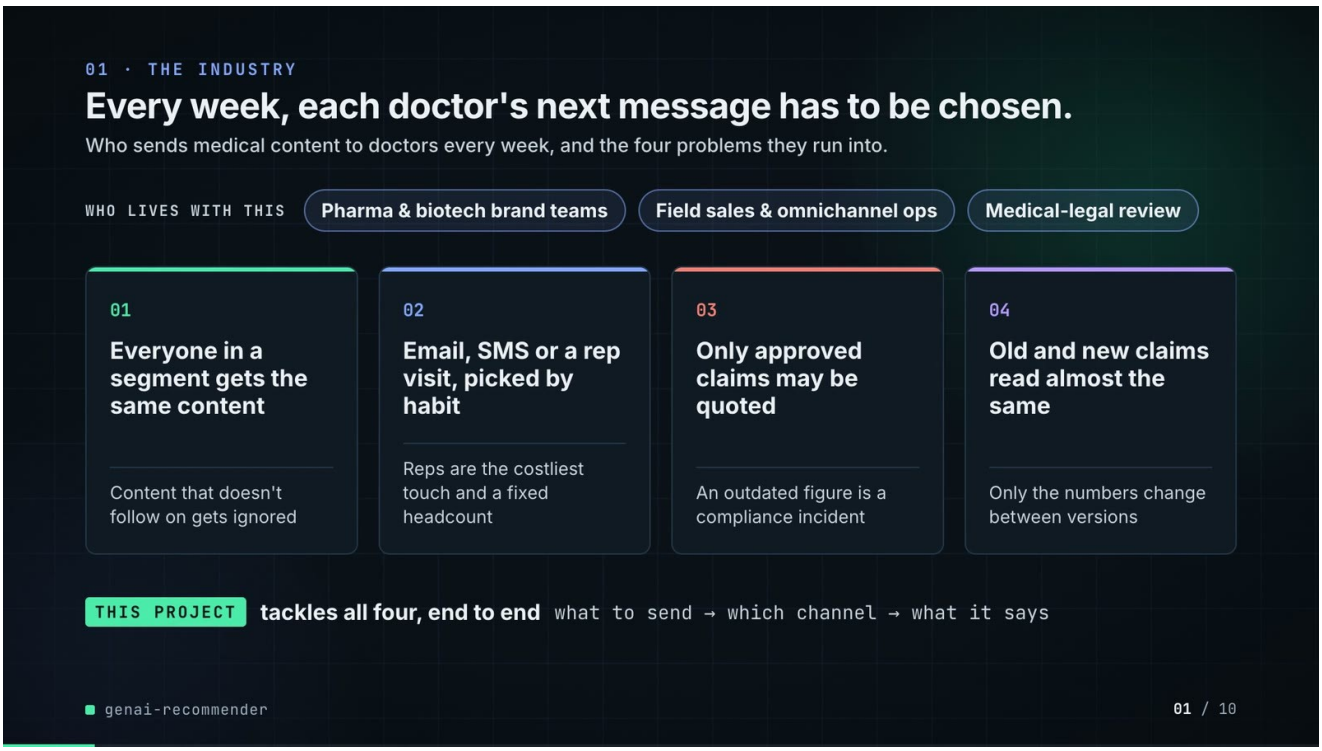
ID	Assumption	Source	Figures affected	Slides affected
A-01	All data is synthetic, produced by a seeded generator sized to the production problem (20,000 prescribers, 480 items, two years, 489 claim snippets). Brand names are invented; the production data is proprietary.	src/genai_recommender/datagen.py:70-76 (Scale), :455 (seed=7), :66 (DAYS = 730); README 'The data'	all figures	slides 01, 02, 03, 04, 05, 06, 07, 08, 09, 10
A-02	Channel policies are scored with the generator's true engagement probability for the channel each policy picks, on the same test touches (last 20% of sends by time). Real logs would need off-policy estimation or a randomised holdout.	src/genai_recommender/channel.py:267-321 (_policy_eval), :163-169 (time_split 70/10/20); README 'Known limitations'	F-50, F-51, F-52, F-54, F-57	slides 08
A-03	Rep visits are a fixed budget equal to the logged (historical) number of rep sends in the test window (13.3% of touches); visits go to the touches where a rep beats the best eligible digital channel by the widest predicted margin. The slide 07 export applies the same share to one region.	src/genai_recommender/channel.py:187-212 (assign_with_rep_budget), :297 (rep_budget); demo tools/export_outreach_plan.py (rep_share)	F-45, F-50, F-51, F-60	slides 07, 08
A-04	Latency is measured on one laptop, 300 requests one at a time, one thread, stub drafter, no server or network hop; p50/p95 vary with machine load.	src/genai_recommender/cli.py:115-140 (request loop, threadpool_limits(limits=1), --latency-n 300); README 'Latency of the request path'	F-17, F-18, F-19, F-20, F-29, F-55, F-56, F-58	slides 05, 08
A-05	The llama3.2 result is three demonstration drafts (3B model, CPU through Ollama, temperature 0.2), not a measured pass rate.	src/genai_recommender/rag.py:122-138 (OllamaLLM); cli.py:70 (--demo-n 3); README 'Latency of the request path'	F-28, F-29, F-30, F-40, F-42, F-43, F-70	slides 05, 06, 10
A-06	The grounding check only checks numbers: every number in a draft must appear in a retrieved approved claim or the content title. Prose that misstates a claim passes (README 'Known limitations', biggest gap).	src/genai_recommender/rag.py:163-174 (grounding_check)	F-10, F-14, F-42, F-43, F-46	slides 03, 04, 06, 07
A-07	Claim retrieval is scored on 79 brand-topic queries with k = 3; the right claims are known by construction (approved snippets written for that brand and topic).	src/genai_recommender/rag.py:64-104 (retrieval_eval), :53 (k = 3)	F-13, F-15, F-16, F-53	slides 04, 08
A-08	Ranking uses a leave-last-out split (train on all but the	src/genai_recommender/	F-22, F-27, F-39, F-61, F-62,	slides 05, 06,

ID	Assumption	Source	Figures affected	Slides affected
	last two reads, validate on the second-to-last, test on the last) after dropping prescribers with more than 40 reads in a day (link scanners). All rankers are scored over the full 480-item catalogue with seen items masked.	`data.py:12 (MAX_EVENTS_PER_DAY = 40), :62-84 (make_split); ranking.py:50-54`	F-64, F-65, F-66, F-72	08, 09
A-09	BERT4Rec configuration: 2 layers, 64 dimensions, 2 heads, 30-item window, up to 30 epochs evaluated every 2, early stop with patience 3 on validation NDCG@10 over 3,000 sampled prescribers, seed 7.	`src/genai_recommender/bert4rec.py:19-35 (Config); ranking.py:62 (valid_users = 3000)`	F-25, F-33, F-34, F-35, F-36, F-38	slides 05, 06
A-10	Slide 07 plans one touch for every prescriber in the north region at the same point in time the request-path test uses (history before the held-out read), with the deterministic stub drafter. The export retrains every model with the repo's seeds; its ranking table reproduces outputs/ranking.csv exactly.	`demo tools/export_outreach_plan.py; src/genai_recommender/cli.py:115-130 (request)`	F-44, F-45, F-46, F-47, F-48, F-49	slides 07
A-11	Run times depend on the machine: the README run is an Apple M1 Pro laptop on 4 CPU threads, no GPU.	`README 'The result' line 57; outputs/summary.json wall_seconds, bert4rec.train_seconds`	F-36, F-37, F-71	slides 06, 10
A-12	Video rounding: probabilities and NDCG to 3 decimals (2 on the slide 07 sheet), shares to 1 decimal or whole percent, latencies to 2 decimals (1 on slide 08), counts with thousands separators; '626k', '300k', '133k', '~4 min' are rounded labels.	`this audit (display convention)`	F-13, F-21, F-25, F-28, F-29, F-47, F-50, F-52, F-55, F-56, F-71	slides 04, 05, 06, 07, 08, 10
A-13	Slide 01 (industry, everyday problems) is general pharma-commercial knowledge used as context, backed by the README problem statement; it carries no numbers.	`author's framing; README 'The problem'`	F-01, F-02	slides 01
A-14	Slide 09's 'payoff, at most' is an upper bound, not a result: it gives each kind of next step (series, resuming a series, format pathway, free choice) whichever of BERT4Rec and the Markov chain scored higher on it, weighted by prescribers. The step labels come from the generator's truth/ folder, which a real blend could not see, and no blend exists in the code (README 'Insights and learning': the blend is the next experiment).	`outputs/ranking_by_segment.csv (step=* slices); README 'Insights and learning', 'Known limitations'`	F-72, F-73	slides 09

4. Slide by slide

Slide 01: The industry

0.0-7.2 s · composition `compositions/s01-context.html`



On-screen text (verbatim)

01 · THE INDUSTRY
Every week, each doctor's next message has to be chosen.
Who sends medical content to doctors every week, and the four problems they run into.
WHO LIVES WITH THIS · Pharma & biotech brand teams · Field sales & omnichannel ops · Medical-legal review
01 Everyone in a segment gets the same content — Content that doesn't follow on gets ignored
02 Email, SMS or a rep visit, picked by habit — Reps are the costliest touch and a fixed headcount
03 Only approved claims may be quoted — An outdated figure is a compliance incident
04 Old and new claims read almost the same — Only the numbers change between versions
THIS PROJECT tackles all four, end to end · what to send → which channel → what it says

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-01	3 chips	`author's framing` · -	-	manual
F-02	4 problem cards	`README 'The problem'` · README lines 13-27	-	✓

Assumptions: A-13

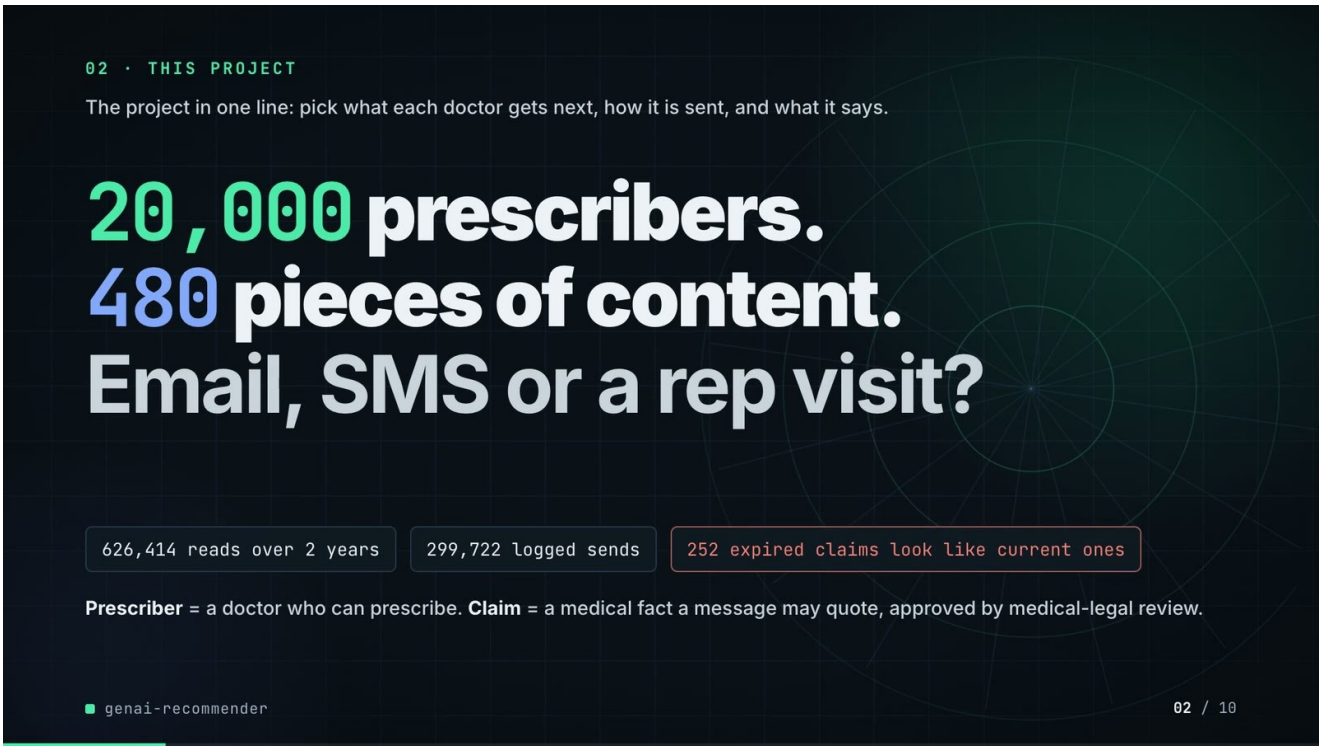
Illustrative, not data:

- The whole slide is context; no repo figures.

To change this slide: Edit the chips (#ctx-c1..3) and cards (#ctx-k1..4) in compositions/s01-context.html.

Slide 02: This project

6.8-12.0 s · composition `compositions/s02-hook.html`



On-screen text (verbatim)

02 · THIS PROJECT

The project in one line: pick what each doctor gets next, how it is sent, and what it says.

20,000 prescribers. 480 pieces of content. Email, SMS or a rep visit?

626,414 reads over 2 years · 299,722 logged sends · 252 expired claims look like current ones

Prescriber = a doctor who can prescribe. Claim = a medical fact a message may quote, approved by medical-legal review.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	20,000 prescribers	`data/customers.csv` · row count	-	✓
F-04	480 pieces of content / top-10 of 480 / all 480 items	`data/items.csv` · row count	-	✓
F-05	Email, SMS or a rep visit?	`src/genai_recommender/datagen.py:62` · CHANNELS	-	✓
F-06	626,414 reads	`data/interactions.csv.gz` · row count	-	✓
F-07	over 2 years / 2 yrs	`src/genai_recommender/datagen.py:65-66`; `data/interactions.csv.gz ts` · DAYS = 730; first and last read 2024-09-01 .. 2026-08-31	span = 729 days + 1	✓
F-08	299,722 logged sends	`data/sends.csv.gz` · row count	-	✓
F-09	252 expired claims look like current ones	`data/snippets.csv` · status == expired	count	✓

Assumptions: A-01

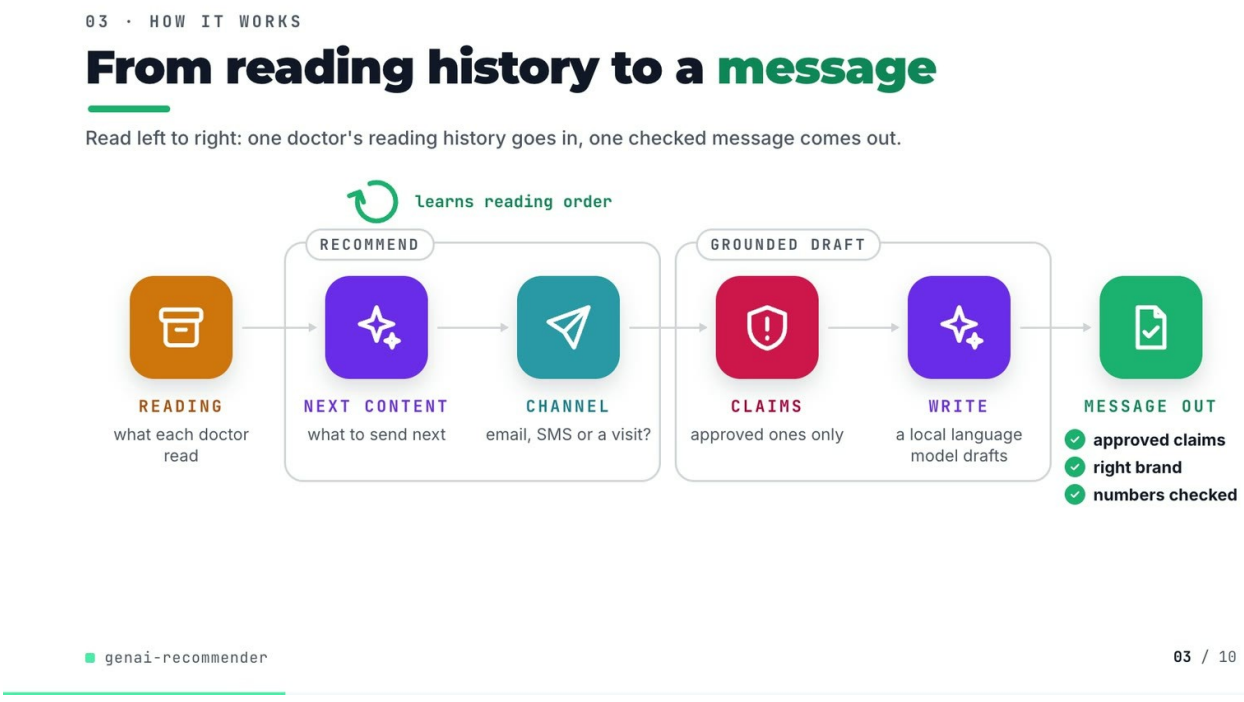
Illustrative, not data:

- Rings in the background are decoration.

To change this slide: Headline spans #hook-l1..l3 and chips #hook-f1..f3 in compositions/s02-hook.html; counts come from data/ (datagen.py Scale).

Slide 03: How it works (white chapter, slide A)

11.7-19.7 s · composition `compositions/explainer.html (CONFIG.blocks, CONFIG.groups, CONFIG.loop)`



On-screen text (verbatim)

03 · HOW IT WORKS

From reading history to a message

Read left to right: one doctor's reading history goes in, one checked message comes out.

READING what each doctor read → [RECOMMEND: NEXT CONTENT what to send next → CHANNEL email, SMS or a visit?] → [GROUNDED DRAFT: CLAIMS approved ones only → WRITE a local language model drafts] → MESSAGE OUT ✓ approved claims ✓ right brand ✓ numbers checked

learns reading order (loop over NEXT CONTENT)

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-05	Email, SMS or a rep visit?	`src/genai_recommender/datagen.py:62` · CHANNELS	-	✓
F-10	3 green badges	`outputs/retrieval.csv; outputs/summary.json` · arm 'approved only + brand filter': unapproved = 0, other-brand = 0; stub_grounded_share = 1.0; llm_demo grounded 3/3	-	✓
F-11	learns reading order	`src/genai_recommender/bert4rec.py:105-130` · _training_batch masking	-	✓

Assumptions: A-06, A-07

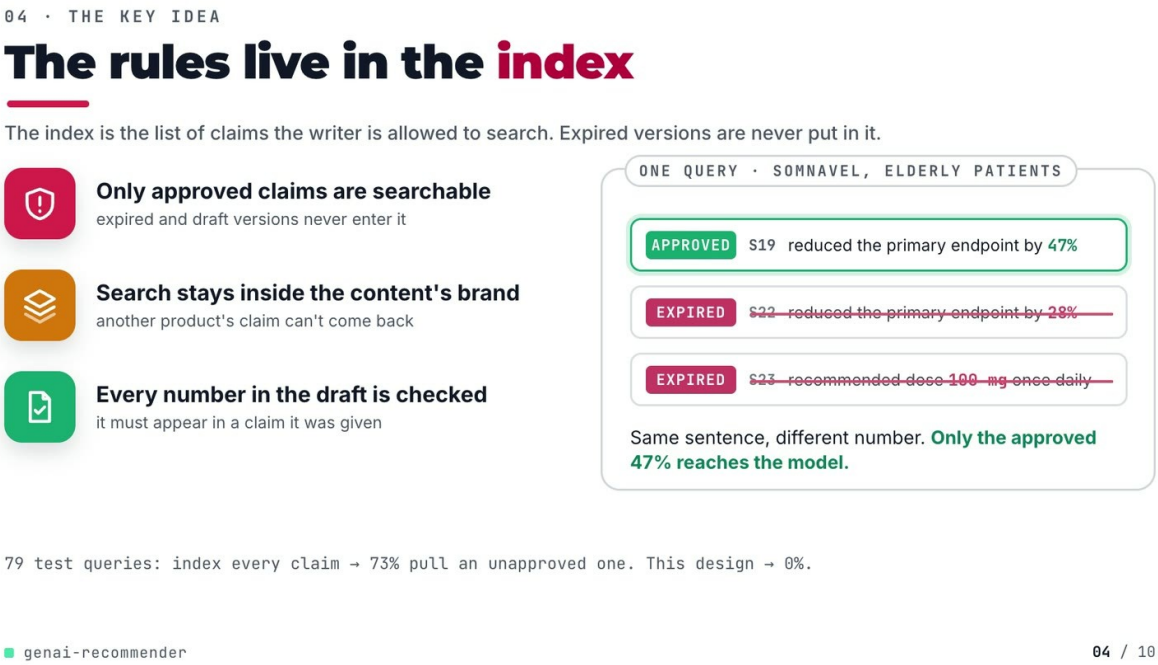
Illustrative, not data:

- Icons and colours are a visual language, not data.
- 'numbers checked' means the number-only grounding check (A-06).

To change this slide: Edit CONFIG in compositions/explainer.html; badge truth comes from retrieval.csv and summary.json (F-10).

Slide 04: The key idea (white chapter, slide B)

19.7-26.7 s · composition `compositions/explainer.html (#ex-b ...)`



On-screen text (verbatim)

04 · THE KEY IDEA

The rules live in the index

The index is the list of claims the writer is allowed to search. Expired versions are never put in it.

Only approved claims are searchable — expired and draft versions never enter it

Search stays inside the content's brand — another product's claim can't come back

Every number in the draft is checked — it must appear in a claim it was given

ONE QUERY · SOMNAVEL, ELDERLY PATIENTS: APPROVED S19 reduced the primary endpoint by 47% · EXPIRED S22 reduced the primary endpoint by 28% (struck) · EXPIRED S23 recommended dose 100 mg once daily (struck)

Same sentence, different number. Only the approved 47% reaches the model.

79 test queries: index every claim → 73% pull an unapproved one. This design → 0%.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-12	S19 APPROVED 47% · S22 EXPIRED 28% · S23 EXPIRED 100 mg	`data/snippets.csv` · snippet_id 19, 22, 23	-	✓
F-13	73%	`outputs/retrieval.csv` · arm='all statuses indexed', any unapproved claim retrieved (0.7342)	x 100, rounded	✓
F-14	3 step statements	`src/genai_recommender/rag.py:36-37, :57-58, :163` · -	-	✓
F-15	This design → 0%	`outputs/retrieval.csv` · arm='approved only + brand filter', any unapproved claim retrieved	x 100	✓
F-16	79 test queries / 79 claim queries	`outputs/retrieval.csv` · queries	-	✓

Assumptions: A-06, A-07, A-12

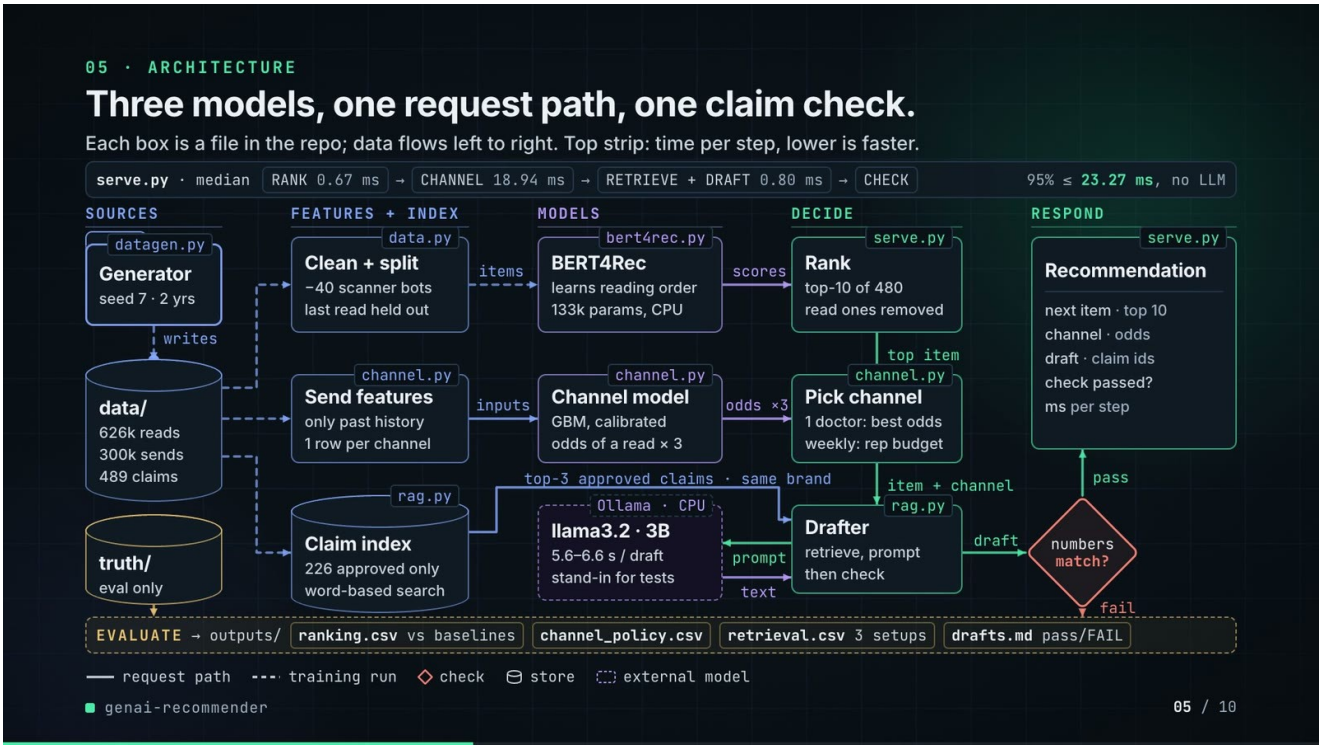
Illustrative, not data:

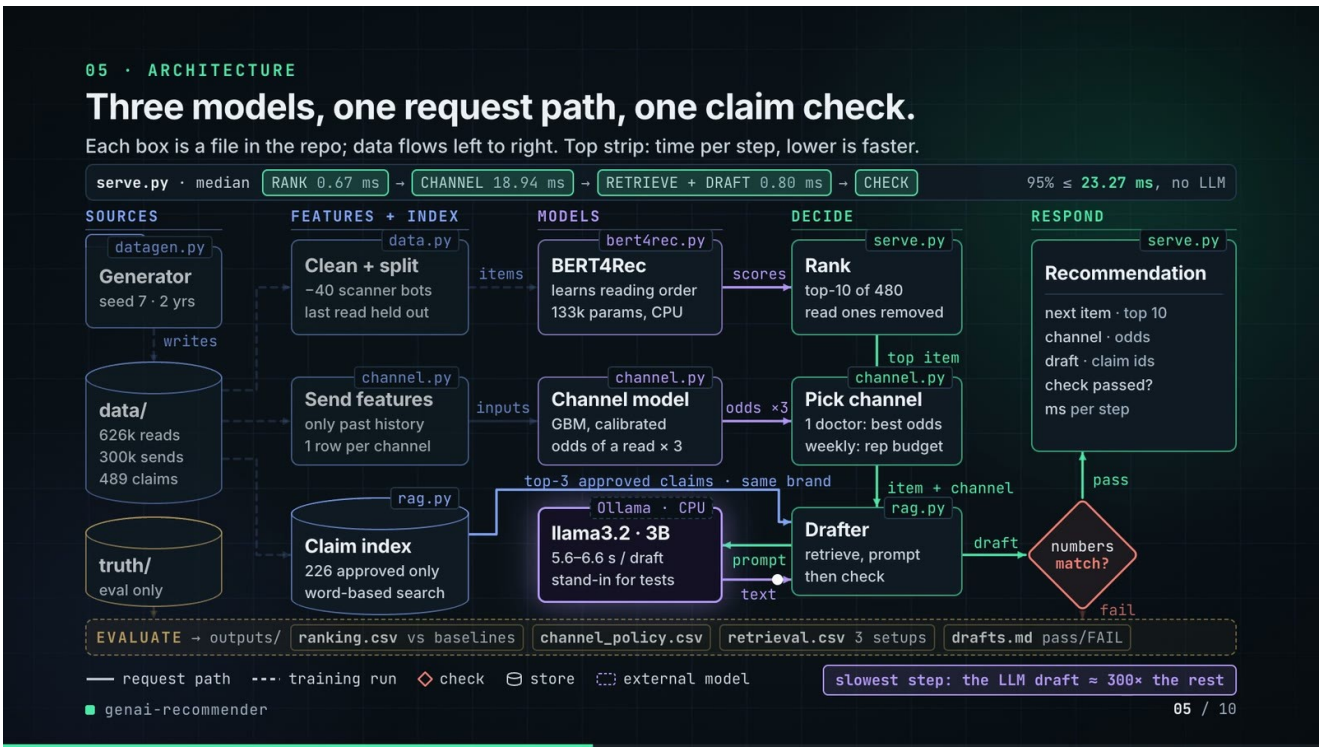
- The three claim cards are real snippets (S19, S22, S23) shortened to their claim; the strike-out animation illustrates that expired claims are never indexed (they are filtered before embedding, not removed at query time); the struck rows stay at 80% opacity so they remain readable.
- The grounding check also accepts numbers from the content title (A-06).

To change this slide: Markup under #ex-b in compositions/explainer.html; the footnote figures are F-13, F-15, F-16.

Slide 05: Architecture

26.4-41.0 s · composition `compositions/s03-architecture.html`





On-screen text (verbatim)

05 · ARCHITECTURE

Three models, one request path, one claim check.

Each box is a file in the repo; data flows left to right. Top strip: time per step, lower is faster.

serve.py · median: RANK 0.67 ms → CHANNEL 18.94 ms → RETRIEVE + DRAFT 0.80 ms → CHECK · 95% ≤ 23.27 ms, no LLM

SOURCES: datagen.py Generator (seed 7 · 2 yrs) → writes → data/ (626k reads, 300k sends, 489 claims); truth/ (eval only, hidden from the models)

FEATURES + INDEX: data.py Clean + split (−40 scanner bots, last read held out); channel.py Send features (only past history, 1 row per channel); rag.py Claim index (226 approved only, word-based search)

MODELS: bert4rec.py BERT4Rec (learns reading order, 133k params, CPU); channel.py Channel model (GBM, calibrated, odds of a read × 3); Ollama · CPU llama3.2 · 3B (5.6–6.6 s / draft, stand-in for tests)

DECIDE: serve.py Rank (top-10 of 480, read ones removed); channel.py Pick channel (1 doctor: best odds, weekly: rep budget); rag.py Drafter (retrieve, prompt, then check)

RESPOND: numbers match? → pass → serve.py Recommendation (next item · top 10, channel · odds, draft · claim ids, check passed?, ms per step); fail → drafts.md

Edge labels: items, scores, inputs, odds ×3, top item, item + channel, top-3 approved claims · same brand, prompt, text, draft, pass, fail, writes

EVALUATE → outputs/ · ranking.csv vs baselines · channel_policy.csv · retrieval.csv 3 setups · drafts.md pass/FAIL

Legend: request path · training run · check · store · external model

slowest step: the LLM draft ≈ 300× the rest

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-04	480 pieces of content / top-10 of 480 / all 480 items	`data/items.csv` · row count	-	✓
F-07	over 2 years / 2 yrs	`src/genai_recommender/datagen.py:65-66`; `data/interactions.csv.gz ts` · DAYS = 730; first and last read 2024-09-01 .. 2026-08-31	span = 729 days + 1	✓
F-17	RANK 0.67 ms	`outputs/latency.csv` · stage=rank, p50 ms	-	✓
F-18	CHANNEL 18.94 ms	`outputs/latency.csv` · stage=channel, p50 ms	-	✓
F-19	RETRIEVE + DRAFT 0.80 ms	`outputs/latency.csv` · stage=retrieve+draft, p50 ms	-	✓
F-20	95% ≤ 23.27 ms, no LLM / 23.3 ms	`outputs/latency.csv` · stage=total, p95 ms	-	✓
F-21	626k reads · 300k sends · 489 claims	`data/interactions.csv.gz, data/sends.csv.gz, data/snippets.csv` · row counts 626,414 / 299,722 / 489	rounded to thousands	✓
F-22	−40 scanner bots	`outputs/summary.json` · scanners_dropped	-	✓
F-23	226 approved only	`data/snippets.csv` · status == approved	count	✓
F-24	word-based search (TF-IDF + SVD, 64-d, not shown in full)	`src/genai_recommender/rag.py:32-44` · SnippetIndex(dim=64)	-	✓
F-25	133k params, CPU (05) · parameters · 2 layers (06)	`outputs/summary.json` · bert4rec.config.layers, dim; bert4rec.parameters = 133,345	parameters rounded to thousands	✓
F-26	seed 7 · 2 yrs	`src/genai_recommender/datagen.py:455, :66` · seed: int = 7; DAYS = 730	-	✓
F-27	top-10 of 480 · read ones removed	`src/genai_recommender/serve.py:60-61` · mask_seen, argsort[:10]	-	✓
F-28	5.6–6.6 s / draft	`outputs/summary.json` · llm_demo.draft_ms = [5570, 6178, 6605]	min, max / 1000	✓
F-29	LLM draft ≈ 300× the rest	`outputs/summary.json; README line 149` · mean draft_ms 6,118 / total p50 20.44 ms = 299	mean(draft_ms) / total p50, rounded to the nearest hundred	✓
F-30	llama3.2 · 3B · Ollama · CPU · stand-in for tests	`README line 148; src/genai_recommender/rag.py StubLLM, OllamaLLM` · -	-	✓
F-31	datagen.py, data.py, channel.py (x3), rag.py (x2), bert4rec.py, serve.py (x2), cli.py, baselines.py	`src/genai_recommender/` · functions and classes	-	✓
F-32	only past history · 1 row per channel	`src/genai_recommender/serve.py:64-72; channel.py:46-58` · -	-	✓

F-29 note: The README states 'roughly 300 times the rest of the request path combined'.

Assumptions: A-01, A-04, A-05, A-06, A-08, A-09

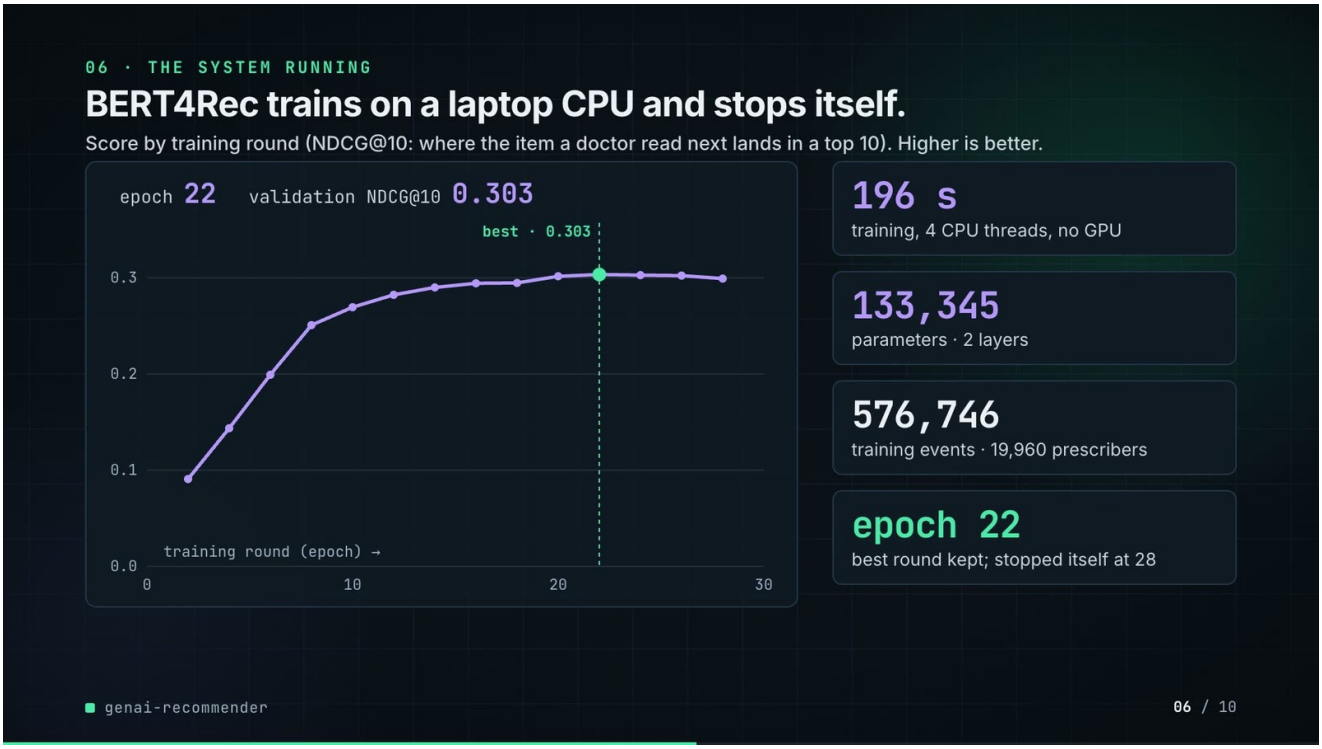
Illustrative, not data:

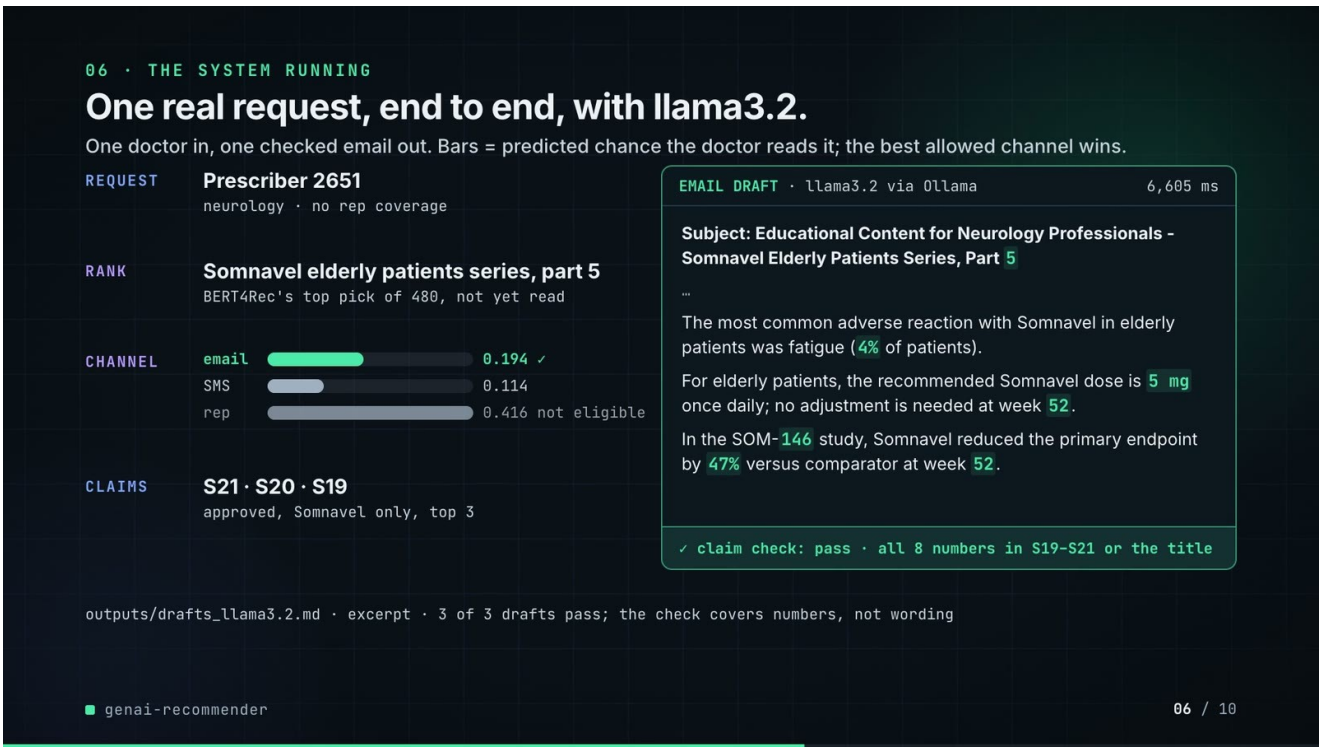
- Packets moving along the wires show execution order, not timing.
- Solid wires = the per-request path in serve.py; dashed (legend: training run) = batch work done once per run (generate, split, train, fit, score).
- 'word-based search' is the TF-IDF + SVD (64-d) claim embedding (F-24); 'GBM, calibrated' is the GBM with isotonic recalibration; 'odds' are the model's predicted engagement probabilities; '95% ≤ 23.27 ms' is the p95 total with the stub drafter.
- Nodes outside the critical path dim to 75% opacity in the last phase (kept readable).
- The fail branch: grounding_check marks the draft grounded=false with the ungrounded numbers, and cli.py writes it as FAIL in drafts_*.md; there is no automatic retry or human approval step in the repo (README 'Known limitations').
- Pick channel takes the eligible argmax per request; the rep budget applies only in the batch policy (README 'How it works').

To change this slide: NODES and the W (wires) object in compositions/s03-architecture.html; band numbers from outputs/latency.csv.

Slide 06: The system running

40.6-55.2 s · composition `compositions/s04-routes.html (HIST array, request trace markup)`





On-screen text (verbatim)

06 · THE SYSTEM RUNNING

BERT4Rec trains on a laptop CPU and stops itself.

Score by training round (NDCG@10: where the item a doctor read next lands in a top 10). Higher is better.

epoch 22 · validation NDCG@10 0.303 (curve of 14 evaluation points, best · 0.303 marker, axis: training round (epoch))

196 s training, 4 CPU threads, no GPU · 133,345 parameters · 2 layers · 576,746 training events · 19,960 prescribers · epoch 22 best round kept; stopped itself at 28

One real request, end to end, with llama3.2.

One doctor in, one checked email out. Bars = predicted chance the doctor reads it; the best allowed channel wins.

REQUEST Prescriber 2651 (neurology · no rep coverage) · RANK Somnavel elderly patients series, part 5 (BERT4Rec's top pick of 480, not yet read) · CHANNEL email 0.194 ✓, SMS 0.114, rep 0.416 not eligible · CLAIMS S21 · S20 · S19 (approved, Somnavel only, top 3)

EMAIL DRAFT · llama3.2 via Ollama · 6,605 ms — Subject line + 3 claim sentences (excerpt) — ✓ claim check: pass · all 8 numbers in S19–S21 or the title

outputs/drafts_llama3.2.md · excerpt · 3 of 3 drafts pass; the check covers numbers, not wording

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-25	133k params, CPU (05) · parameters · 2 layers (06)	`outputs/summary.json` · bert4rec.config.layers, dim; bert4rec.parameters = 133,345	parameters rounded to thousands	✓
F-33	14 points, epochs 2-28	`outputs/bert4rec_training.json history` · epoch, valid_ndcg10	-	✓
F-34	epoch 22 · 0.303	`outputs/bert4rec_training.json` · best_epoch, best_valid_ndcg10	-	✓
F-35	stopped at 28	`outputs/bert4rec_training.json` · last history epoch	-	✓
F-36	196 s	`outputs/summary.json` · bert4rec.train_seconds	-	✓
F-37	4 CPU threads, no GPU	`README line 57; cli.py --threads default min(4, cpu_count)` · -	-	✓
F-38	133,345	`outputs/summary.json` · bert4rec.parameters	-	✓
F-39	576,746 training events · 19,960 prescribers	`outputs/summary.json` · split.train_events, split.users	-	✓
F-40	Prescriber 2651 · Somnavel elderly patients series, part 5 (BERT4Rec's top pick of 480, not yet read) · email 0.194 / SMS 0.114 / rep 0.416 · S21 S20 S19 · 6,605 ms	`outputs/drafts_llama3.2.md lines 44-49` · Customer 2651 section	-	✓
F-41	neurology · no rep coverage · rep 0.416 not eligible	`data/customers.csv` · customer_id 2651	-	✓
F-42	claim check: pass · all 8 numbers in S19–S21 or the title	`outputs/drafts_llama3.2.md; data/snippets.csv; rag.py grounding_check logic` · -	numbers in draft (regex \d+(\.\d+)?) all in claims + title	✓
F-43	3 of 3 drafts pass	`outputs/summary.json` · llm_demo.samples, grounded	-	✓

Assumptions: A-05, A-06, A-09, A-11

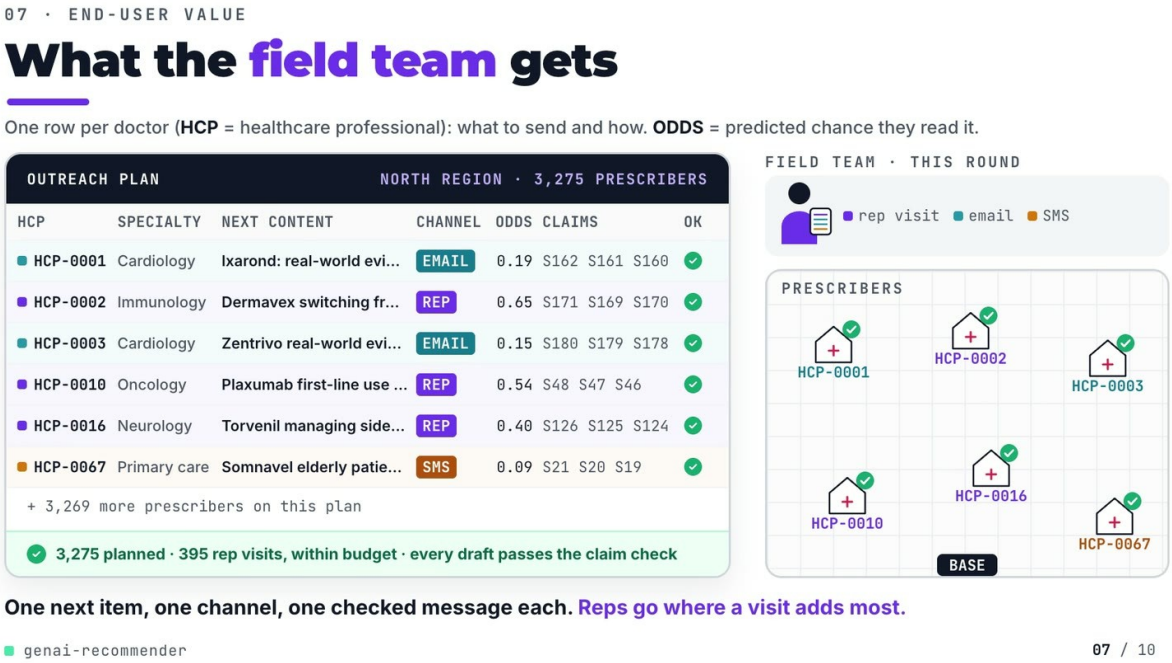
Illustrative, not data:

- The curve is drawn from bert4rec_training.json (F-33); the readout steps through the logged points.
- The draft card shows the subject and the three claim sentences of the real draft; the greeting and closing lines are elided ('…').
- The latency counter animates up to the logged 6,605 ms; it is not a live timer.

To change this slide: HIST array and trace markup in compositions/s04-routes.html; the trace is the Customer 2651 section of outputs/drafts_llama3.2.md.

Slide 07: End-user value (white)

54.9-67.9 s · composition `compositions/value.html` (SHEET block injected by `tools/inject_sheet.py`)



On-screen text (verbatim)

07 · END-USER VALUE

What the field team gets

One row per doctor (HCP = healthcare professional): what to send and how. ODDS = predicted chance they read it.

OUTREACH PLAN · NORTH REGION · 3,275 PRESCRIBERS — columns HCP, SPECIALTY, NEXT CONTENT, CHANNEL, ODDS, CLAIMS, OK

HCP-0001 Cardiology · Ixarond: real-world evidence (article) · EMAIL 0.19 · S162 S161 S160 ✓

HCP-0002 Immunology · Dermavex switching from prior therapy series, part … · REP 0.65 · S171 S169 S170 ✓

HCP-0003 Cardiology · Zentrivo real-world evidence series, part 5 · EMAIL 0.15 · S180 S179 S178 ✓

HCP-0010 Oncology · Plaxumab first-line use series, part 4 · REP 0.54 · S48 S47 S46 ✓

HCP-0016 Neurology · Torvenil managing side effects series, part 1 · REP 0.40 · S126 S125 S124 ✓

HCP-0067 Primary care · Somnavel elderly patients series, part 4 · SMS 0.09 · S21 S20 S19 ✓

+ 3,269 more prescribers on this plan · ✓ 3,275 planned · 395 rep visits, within budget · every draft passes the claim check

FIELD TEAM · THIS ROUND · rep visit / email / SMS · PRESCRIBERS · BASE

One next item, one channel, one checked message each. Reps go where a visit adds most.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-44	3,275 prescribers / 3,275 planned	`demo assets/outreach_plan.json` · prescribers	-	✓
F-45	395 rep visits, within budget	`demo assets/outreach_plan.json` · channels.rep = 395, rep_budget = 435	395 <= 435	✓
F-46	every draft passes the claim check	`demo assets/outreach_plan.json` · grounded == prescribers	3,275 / 3,275	✓
F-47	HCP-0001, -0002, -0003, -0010, -0016, -0067	`demo assets/outreach_plan.json, injected into compositions/value.html` · rows 0, 1, 2, 9, 15, 66	HCP id = row position + 1	✓
F-48	+ 3,269 more prescribers	F-44 · 3,275 - 6	-	✓
F-49	(supports every slide 07 row)	`demo assets/outreach_plan.json ranking_check vs outputs/ranking.csv` · HR@10, NDCG@10, coverage@10 per model	equal at 4 decimals	✓
F-60	(rep budget, slide 07 'within budget')	`outputs/channel_policy.csv; demo assets/outreach_plan.json` · rep share of the logged policy = 0.1328 = export rep_share_historical	-	✓

Assumptions: A-03, A-06, A-10, A-12

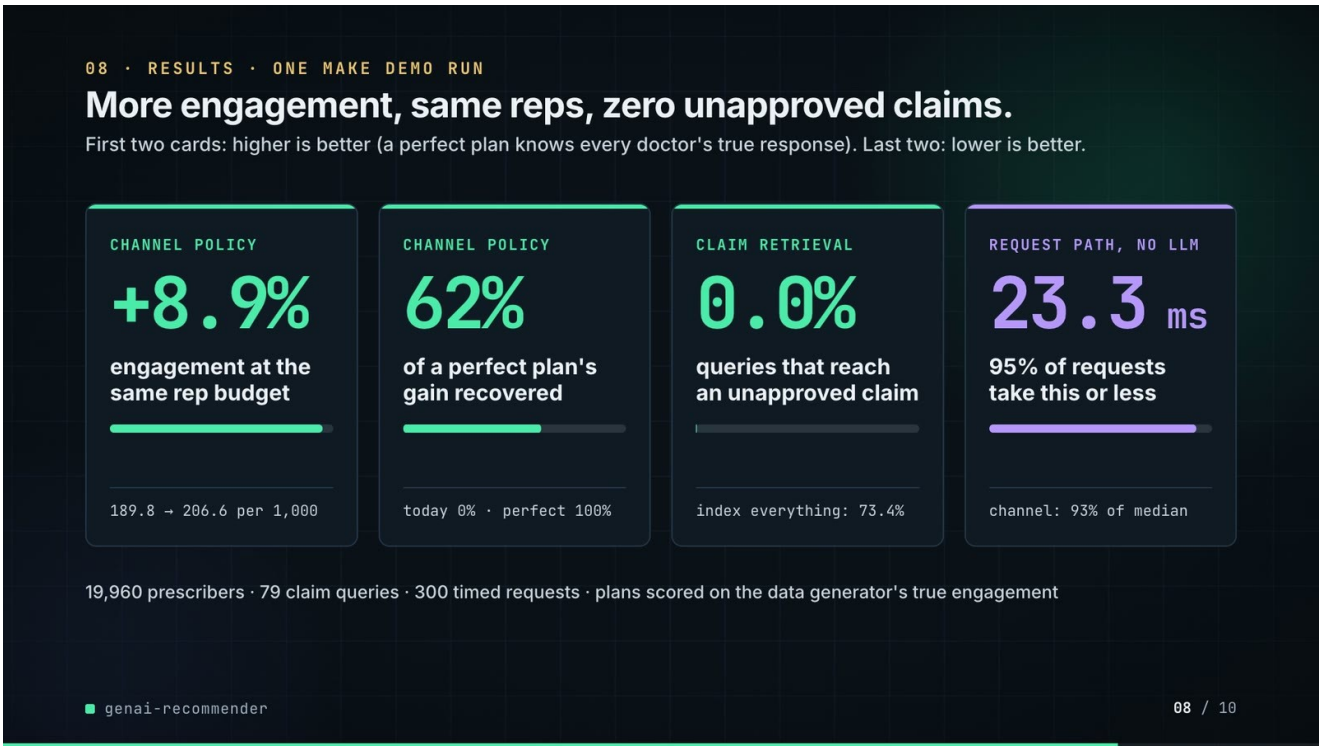
Illustrative, not data:

- HCP-xxxx are row positions in the export, not real people.
- Content titles are truncated on screen to fit the column.
- Clinic positions, the base, the rep figure, envelopes, SMS bubbles and the car are illustrative; the data has no geography.
- Green checks on clinics mean the claim-checked message was planned for that prescriber, not that the prescriber engaged.
- 'Reps go where a visit adds most' paraphrases the rep-budget rule (largest predicted rep-over-digital margin, A-03).

To change this slide: Re-run tools/export_outreach_plan.py, then tools/inject_sheet.py (--rows picks the rows); layout in compositions/value.html.

Slide 08: Results

67.6-76.4 s · composition `compositions/s05-results.html (stats array)`



On-screen text (verbatim)

08 · RESULTS · ONE MAKE DEMO RUN

More engagement, same reps, zero unapproved claims.

First two cards: higher is better (a perfect plan knows every doctor's true response). Last two: lower is better.

CHANNEL POLICY +8.9% engagement at the same rep budget · 189.8 → 206.6 per 1,000

CHANNEL POLICY 62% of a perfect plan's gain recovered · today 0% · perfect 100%

CLAIM RETRIEVAL 0.0% queries that reach an unapproved claim · index everything: 73.4%

REQUEST PATH, NO LLM 23.3 ms 95% of requests take this or less · channel: 93% of median

19,960 prescribers · 79 claim queries · 300 timed requests · plans scored on the data generator's true engagement

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-16	79 test queries / 79 claim queries	`outputs/retrieval.csv` · queries	-	✓
F-20	95% ≤ 23.27 ms, no LLM / 23.3 ms	`outputs/latency.csv` · stage=total, p95 ms	-	✓
F-39	576,746 training events · 19,960 prescribers	`outputs/summary.json` · split.train_events, split.users	-	✓
F-50	+8.9%	`outputs/channel_policy.csv` · engaged per 1k touches: model 206.64 vs logged 189.77	(206.64 / 189.77 - 1) x 100 = 8.89%	✓
F-51	189.8 → 206.6 per 1,000	`outputs/channel_policy.csv` · engaged per 1k touches	-	✓
F-52	62% of a perfect plan's gain recovered	`outputs/channel_policy.csv` · policy='model, same rep budget', share of oracle lift (0.6173)	x 100	✓
F-53	0.0% (index everything: 73.4%)	`outputs/retrieval.csv` · any unapproved claim retrieved	x 100	✓
F-54	today 0% · perfect 100%	`outputs/channel_policy.csv` · share of oracle lift for logged and oracle	-	✓
F-55	23.3 ms · 95% of requests take this or less · request path, no LLM	`outputs/latency.csv` · stage=total, p95 ms	1 decimal	✓
F-56	channel: 93% of median	`outputs/latency.csv` · 18.94 / 20.44	channel p50 / total p50 x 100 = 92.7%	✓
F-57	plans scored on the data generator's true engagement	`src/genai_recommender/channel.py:274` · docstring	-	✓
F-58	300 timed requests	`outputs/latency.csv` · requests	-	✓
F-59	one make demo run	`README line 57; outputs/summary.json llm_demo present` · -	-	✓
F-60	(rep budget, slide 07 'within budget')	`outputs/channel_policy.csv; demo assets/outreach_plan.json` · rep share of the	-	✓

ID	Shown	Source · key	Calculation	Check
		logged policy = 0.1328 = export rep_share_historical		

Assumptions: A-02, A-03, A-04, A-07, A-12

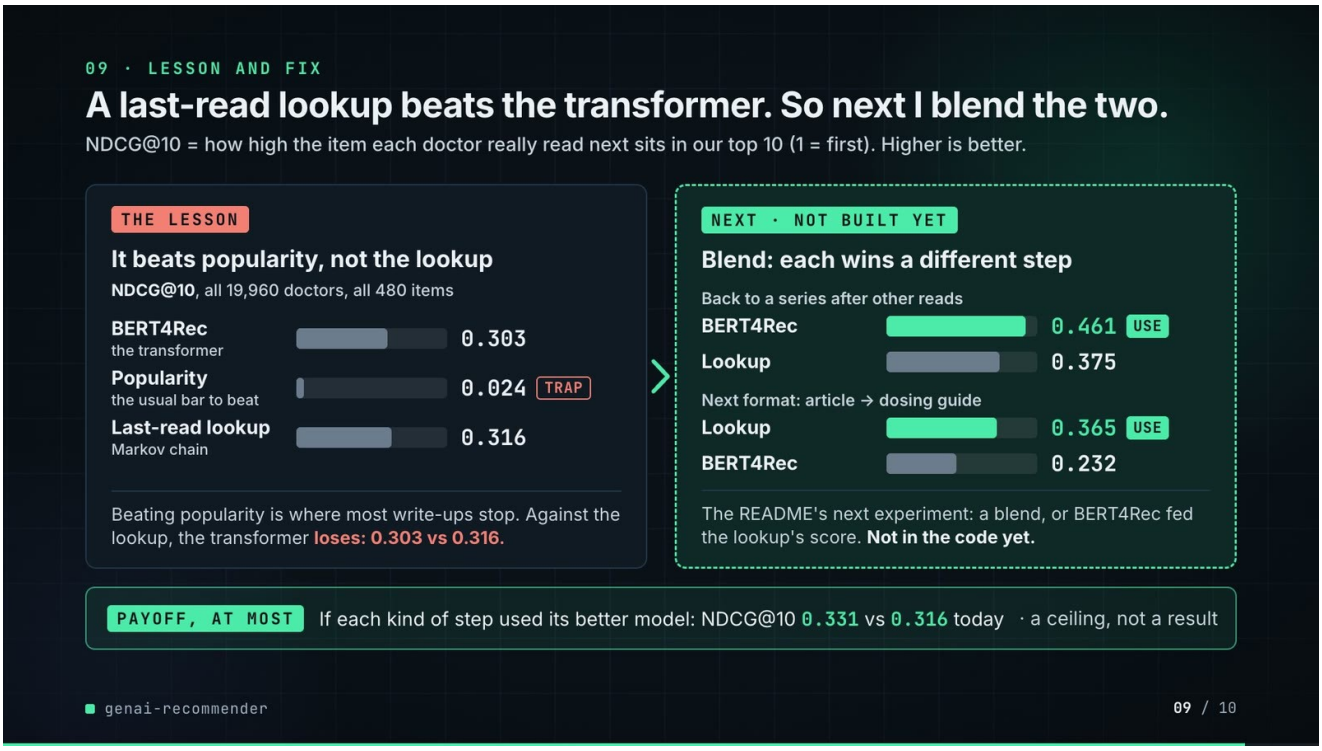
Illustrative, not data:

- 'Perfect plan' = the oracle policy (true engagement known, same rep budget); 'today' = the logged historical policy. The 0.0% card is mint because zero unapproved claims is the good outcome.
- Mini bars: card 1 = engaged per 1,000 as a share of the oracle's 217.1 (logged → model); card 2 = share of oracle lift; card 3 = 73.4% → 0%; card 4 = channel step's share of the p50 request.

To change this slide: The stats array and card text in compositions/s05-results.html.

Slide 09: Lesson and fix

76.0-85.0 s · composition `compositions/s06-twist.html`



On-screen text (verbatim)

09 · LESSON AND FIX
A last-read lookup beats the transformer. So next I blend the two.
NDCG@10 = how high the item each doctor really read next sits in our top 10 (1 = first). Higher is better.
THE LESSON · It beats popularity, not the lookup · NDCG@10, all 19,960 doctors, all 480 items
BERT4Rec (the transformer) 0.303 · Popularity (the usual bar to beat) 0.024 TRAP · Last-read lookup (Markov chain) 0.316
Beating popularity is where most write-ups stop. Against the lookup, the transformer loses: 0.303 vs 0.316.
NEXT · NOT BUILT YET · Blend: each wins a different step
Back to a series after other reads: BERT4Rec 0.461 USE · Lookup 0.375
Next format: article → dosing guide: Lookup 0.365 USE · BERT4Rec 0.232
The README's next experiment: a blend, or BERT4Rec fed the lookup's score. Not in the code yet.
PAYOFF, AT MOST · If each kind of step used its better model: NDCG@10 0.331 vs 0.316 today · a ceiling, not a result

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-04	480 pieces of content / top-10 of 480 / all 480 items	`data/items.csv` · row count	-	✓
F-39	576,746 training events · 19,960 prescribers	`outputs/summary.json` · split.train_events, split.users	-	✓
F-61	0.316 · loses: 0.303 vs 0.316	`outputs/ranking.csv` · model='last-item Markov', NDCG@10	-	✓
F-62	0.303	`outputs/ranking.csv` · model='BERT4Rec', NDCG@10	-	✓
F-64	0.024	`outputs/ranking.csv` · model='popularity', NDCG@10	-	✓
F-65	Lookup 0.365 USE · BERT4Rec 0.232	`outputs/ranking_by_segment.csv` · slice='step=pathway', NDCG@10	-	✓
F-66	BERT4Rec 0.461 USE · Lookup 0.375	`outputs/ranking_by_segment.csv` · slice='step=series_resume', NDCG@10	-	✓
F-72	NDCG@10 0.331 vs 0.316 today · a ceiling, not a result	`outputs/ranking_by_segment.csv` · slices step=series, series_resume, pathway, pref (users 3,624 + 2,392 + 4,145 + 9,799 = 19,960)	sum(users x max(BERT4Rec, Markov)) / 19,960 = 0.3312; Markov alone = 0.3158 (the same weighting reproduces outputs/ranking.csv for both models)	✓
F-73	NEXT · NOT BUILT YET · The README's next experiment: a blend, or BERT4Rec fed the lookup's score. Not in the code yet.	`README 'Insights and learning'; src/genai_recommender/*.py; serve.py:59 · -`	-	✓

ID	Shown	Source · key	Calculation	Check
F-74	Back to a series after other reads · Next format: article → dosing guide	README 'The result' slice table · -	-	✓

F-72 note: Upper bound only (A-14): step labels come from truth/, so a deployable blend cannot route on them.

Assumptions: A-08, A-09, A-12, A-14

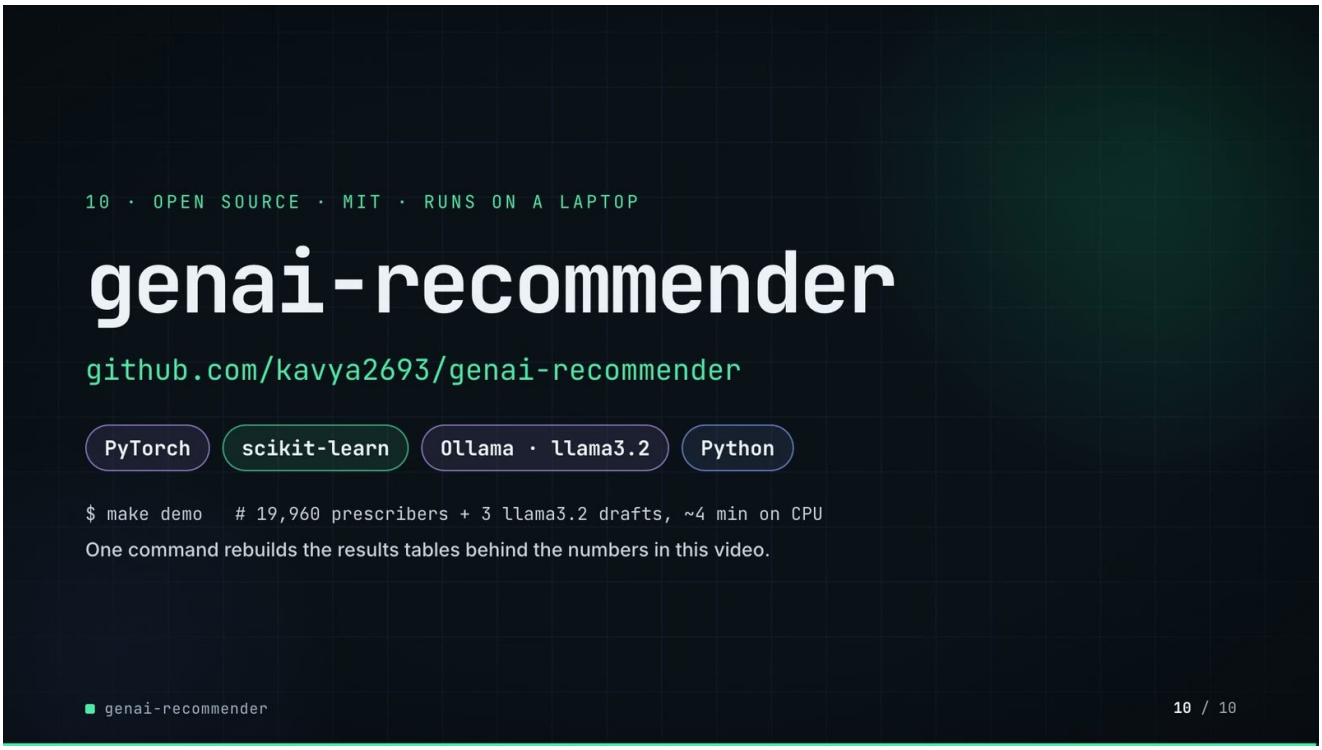
Illustrative, not data:

- Bars share one scale in both panels: 220 px = NDCG@10 0.5. All compared bars are neutral grey; coral marks only the TRAP label (popularity is the misleading comparison) and mint marks the model a blend would use for that kind of step.
- The right panel is labelled NEXT because the repo has no fix built for this result (F-73); USE marks what a blend would pick, not something the code does today.
- 'Next format: article → dosing guide' is the README's own example for the format-pathway slice; the slice covers every format pathway within a brand.
- The payoff strip is an upper bound (A-14), not a measured result.

To change this slide: Panels #twist-left / #twist-right and the payoff strip in compositions/s06-twist.html (bar widths are inline styles on the shared scale).

Slide 10: End card

84.6-89.0 s · composition `compositions/s07-end.html`



On-screen text (verbatim)

10 · OPEN SOURCE · MIT · RUNS ON A LAPTOP
genai-recommender
github.com/kavya2693/genai-recommender
PyTorch · scikit-learn · Ollama · llama3.2 · Python
$ make demo # 19,960 prescribers + 3 llama3.2 drafts, ~4 min on CPU
One command rebuilds the results tables behind the numbers in this video.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-67	MIT	`LICENSE` line 1 · -	-	✓
F-68	PyTorch · scikit-learn · Ollama · llama3.2 · Python	`pyproject.toml; src/genai_recommender/rag.py` · torch, scikit-learn, OllamaLLM(model='llama3.2')	-	✓
F-69	$ make demo	`Makefile` · demo target	-	✓
F-70	+ 3 llama3.2 drafts	`Makefile demo target` · --demo-n 3	-	✓
F-71	~4 min on CPU	`outputs/summary.json` · wall_seconds = 233.4	233.4 / 60 = 3.9 min	✓

Assumptions: A-11, A-12

To change this slide: Text in compositions/s07-end.html.

5. Change-impact matrix

Input / assumption	Affected figures	Slides	Files to edit	Re-run
A-01: All data is synthetic, produced by a seeded generator sized to the production problem (20,000 prescribers, 480 items, two years, 489 claim snippets)	all figures	01, 02, 03, 04, 05, 06, 07, 08, 09, 10	`src/genai_recommender/datagen.py (Scale)`	make data; make demo
A-02: Channel policies are scored with the generator's true engagement probability for the channel each policy picks, on the same test touches (last 20% of sends by time)	F-50, F-51, F-52, F-54, F-57	08	`src/genai_recommender/channel.py`	make demo
A-03: Rep visits are a fixed budget equal to the logged (historical) number of rep sends in the test window (13	F-45, F-50, F-51, F-60	07, 08	`src/genai_recommender/channel.py:297`; `tools/export_outreach_plan.py`	make demo; export_outreach_plan.py + inject_sheet.py
A-04: Latency is measured on one laptop, 300 requests one at a time, one thread, stub drafter, no server or network hop; p50/p95 vary with machine load	F-17, F-18, F-19, F-20, F-29, F-55, F-56, F-58	05, 08	`src/genai_recommender/cli.py:71`	make demo
A-05: The llama3	F-28, F-29, F-30, F-40, F-42, F-43, F-70	05, 06, 10	`Makefile (demo target)`; `src/genai_recommender/rag.py:130`	make demo
A-06: The grounding check only checks numbers: every number in a draft must appear in a retrieved approved claim or the content title	F-10, F-14, F-42, F-43, F-46	03, 04, 06, 07	`src/genai_recommender/rag.py:163-174`	make demo
A-07: Claim retrieval is scored on 79 brand-topic queries with k = 3; the right claims are known by construction (approved snippets written for that brand and topic)	F-13, F-15, F-16, F-53	04, 08	`src/genai_recommender/rag.py`	make demo
A-08: Ranking uses a leave-last-out split (train on all but the last two reads, validate on the second-to-last, test on the last) after dropping prescribers with more than 40 reads in a day (link scanners)	F-22, F-27, F-39, F-61, F-62, F-64, F-65, F-66, F-72	05, 06, 08, 09	`src/genai_recommender/data.py`	make demo
A-09: BERT4Rec configuration: 2 layers, 64 dimensions, 2 heads, 30-item window, up to 30 epochs evaluated every 2, early stop with patience 3 on validation NDCG@10 over 3,000 sampled prescribers, seed 7	F-25, F-33, F-34, F-35, F-36, F-38	05, 06	`src/genai_recommender/bert4rec.py:19-35`	make demo; export_outreach_plan.py + inject_sheet.py
A-10: Slide 07 plans one touch for every prescriber in the north region at the same point in time the request-path test uses (history before the held-out read), with the deterministic stub drafter	F-44, F-45, F-46, F-47, F-48, F-49	07	`tools/export_outreach_plan.py (--region)`; `tools/inject_sheet.py (--rows)`	export_outreach_plan.py + inject_sheet.py; hyperframes render
A-11: Run times depend on the machine: the README run is an Apple M1 Pro laptop on 4 CPU threads, no GPU	F-36, F-37, F-71	06, 10	-	make demo
A-12: Video rounding: probabilities and NDCG to 3 decimals (2 on the slide 07 sheet), shares to 1 decimal or whole percent, latencies to 2 decimals (1 on slide 08), counts with thousands separators; '626k', '300k', '133k', '~4 min' are rounded labels	F-13, F-21, F-25, F-28, F-29, F-47, F-50, F-52, F-55, F-56, F-71	04, 05, 06, 07, 08, 10	`the composition file named on each slide`	hyperframes render
A-13: Slide 01 (industry, everyday problems) is general pharma-commercial knowledge used as context, backed by the README problem statement; it carries no numbers	F-01, F-02	01	`compositions/s01-context.html`	hyperframes render
A-14: Slide 09's 'payoff, at most' is an upper bound, not a result: it gives each kind of next step (series, resuming a series, format pathway, free choice) whichever of BERT4Rec and the Markov chain scored higher on it, weighted by prescribers	F-72, F-73	09	`compositions/s06-twist.html`	make demo; hyperframes render

6. Known differences and caveats

• Channel-policy numbers (+8.9%, 62%) are scored with the generator's own engagement function, which real logs do not provide (A-02).

• The claim check covers numbers only; README 'Known limitations' shows two of the three llama3.2 drafts passing it while misstating a claim's prose. Slide 06 shows the one draft whose prose follows its claims.

• llama3.2 results are three demonstration drafts, not a pass rate (A-05).

• Latency is single-machine, single-thread, stub drafter; the max column (53.82 ms) is first-touch outliers (README).

• The headline sequence result depends on the generator: format pathways are first-order by construction, which favours the Markov chain (README 'Known limitations').

• Slide 07 is one region's plan exported by re-running the repo's pipeline (A-10); HCP ids are row positions, not people.

• Slide 01 statements are general industry knowledge (A-13).

• Slide 09 no longer shows item-kNN (NDCG@10 0.102), the free-choice and same-series slices or the new-content slice; the README 'The result' tables still report all of them.

• Slide 09's payoff strip (0.331) is a ceiling computed with generator step labels a deployed blend could not see; the blend is not built (A-14, F-73).

• The request path in serve.py still ranks with BERT4Rec even though the Markov chain scores higher overall; the README states this and names the blend as the next experiment.

Appendix: environment

Item	Value
Commit	`37d49ecc8dcf2a405ea628ac0a26027e2d2aaa41`
Python	3.11.7
torch	2.9.1
sklearn	1.9.1
pandas	3.0.6
numpy	2.4.6
Profile / seeds	data_generator=7, bert4rec=7, latency_sample_rng=1, validation_users_rng=0
Run time	summary.json wall_seconds = 233.4 s for make demo (README: Apple M1 Pro laptop, 4 threads, no GPU, including three llama3.2 drafts); the slide 07 export took 351.9 s